



MARITIME SECURITY

Threat Detection & Risk Assessment System

Maritime Security Bootcamp





Meet The Team



Kavya Bhalodia ⚓

B.S.
Computer Science &
Data Science



Arnav Nethula ⚓

B.S.
Computer Science
& Data Science



Ayaan Alam ⚓

B.S.
Computer Science
& Data Science



Keerthi Peruka ⚓

B.S.
Computer Science
& Data Science

RUTGERS MBS ADVISOR



Prof. Brian Petrus



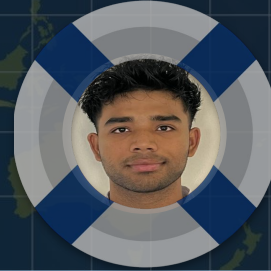
Muhammad Ali ⚓

B.S.
Computer Science &
Data Science



Srisai Varanasi ⚓

B.S.
Computer Science



Rakesh Cavala ⚓

B.S.
Computer Science &
Data Science

MARITIME SECURITY VISUAL AGENDA DASHBOARD

01 — BACKGROUND



- Maritime industry essential for global trade, facing growing security threats & risks
- 12,000+ GPS spoofing records
- Major rise in sanction evasion & identity manipulation



02 — PROBLEM STATEMENT

No unified real-time vessel tracking & risk assessment
Threats: GPS spoofing, identity manipulation, dark activity, national security risks

GPS RISK



IDENTITY



DARK AIS



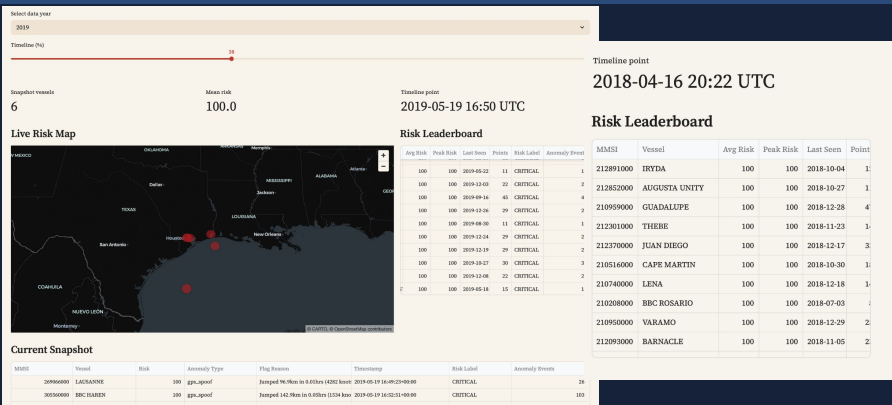
NAT SEC



03 — APPROACH

- 01 DATA COLLECTION — AIS/REGISTRY/SANCTIONS
- 02 PRE-PROCESSING
- 03 GPS ANOMALY DETECTION
- 04 IDENTITY RISK SCORING
- 05 COMBINED RISK SCORE
- 06 VISUAL DASHBOARD

04 — RESULTS



05 — KEY LEARNINGS

- Learned that AIS data reveals behavior registries and sanctions lists miss, and that combining all three sources makes risk detection significantly more accurate
- Learned that risk scoring only works when you define clear triggers, speed over 50 knots, 200+ km location jumps, name changes, flag switches, and dark periods all turned out to be strong indicators of evasion

1: Background

The Maritime Industry: Essential for global trade but facing **growing security risks**:

- **Major rise in** sanction evasion
- Over **12000 records** of GPS spoofing
- **Multiple records** of Identity manipulation



2: Problem Statement / Goals

Statement: The maritime industry faces increased security risks such as vessel identity, GPS spoofing, cyber threats, and national security. It is important to have a better secured monitoring maritime system for detecting vessel tracking and risk assessment methods.

Real-time GPS anomaly
detection (Anomaly risk)

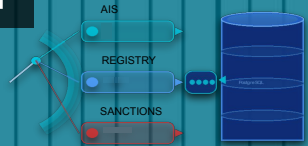
Vessel Identity Risk
Engine (Identity risk)

Combined Risk Scoring
System

3: Approach

Data Collection

01



Pre-Processing

02



GPS Anomaly Detect

03



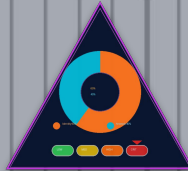
Identity Risk Score

04



Combined Risk Score

05



Dashboard & Alerts

06



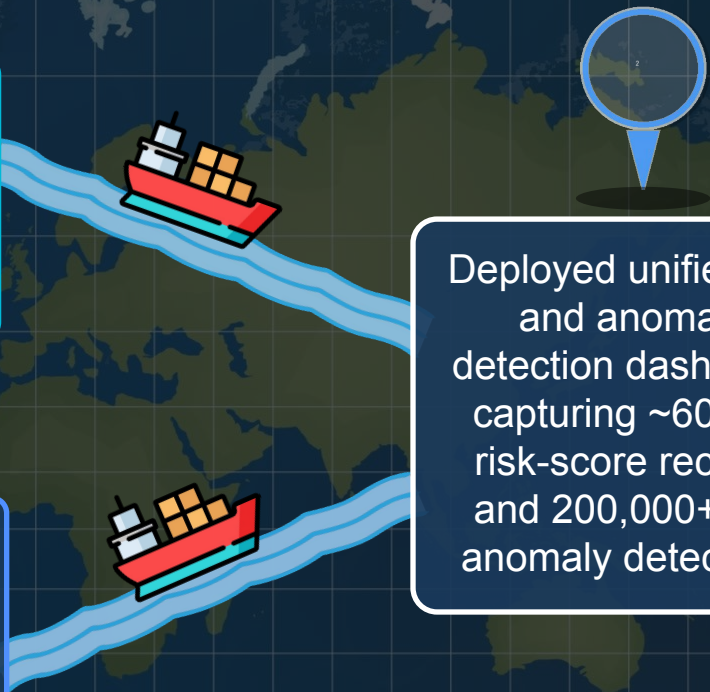
4. Results

Risk Labels:

- **LOW:** 56.8%
- **MEDIUM:** 12.11%
- **HIGH:** 18.41%
- **CRITICAL:** 12.68%

Anomaly Types:

- **Impossible_speed:** 66.32%
- **AIS_dark:** 24.66%
- **GPS_spoof:** 9.02%



Deployed unified risk and anomaly detection dashboard capturing ~60,000 risk-score records and 200,000+ AIS anomaly detections

5: Key Learnings

TAKEAWAYS / UNDERSTANDING

- GPS spoofing & identity risks are more common than expected
- More data sources = better risk assessment

TECHNICAL SKILLS

- Built a Python pipeline (Pandas/NumPy) to process 100+ AIS files, flagging GPS spoofing and dark periods (6+ hr gaps)

BUSINESS SKILLS

- Broke down a real-world problem into a structured solution
- Improved teamwork and communication while working on a technical project

01 — BACKGROUND



- Maritime industry essential for global trade, facing growing security threats & risks
- 12,000+ GPS spoofing records
- Major rise in sanction evasion & identity manipulation



02 — PROBLEM STATEMENT

No unified real-time vessel tracking & risk assessment
Threats: GPS spoofing, identity manipulation, dark activity, national security risks

GPS RISK



IDENTITY



DARK AIS



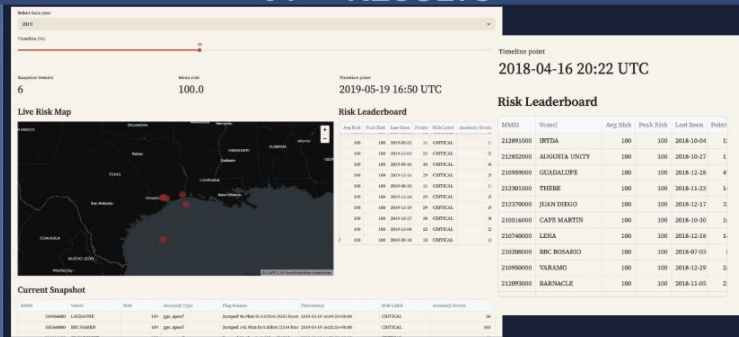
NAT SEC



03 — APPROACH

- 01 DATA COLLECTION — AIS/REGISTRY/SANCTIONS
- 02 PRE-PROCESSING
- 03 GPS ANOMALY DETECTION
- 04 IDENTITY RISK SCORING
- 05 COMBINED RISK SCORE
- 06 VISUAL DASHBOARD

04 — RESULTS



05 — KEY LEARNINGS

- Learned that AIS data reveals behavior registries and sanctions lists miss, and that combining all three sources makes risk detection significantly more accurate
- Learned that risk scoring only works when you define clear triggers, speed over 50 knots, 200+ km location jumps, name changes, flag switches, and dark periods all turned out to be strong indicators of evasion



THANK YOU!

